

Lesson 6: Resistance in combinations

Electric Circuits · Year 10 · 50 minutes · AQA RP 15 / RP 3

Why this lesson is here

Adding a second resistor changes the current drawn from the same supply. The direction of change depends on the wiring.

Learning objective

Measure equivalent resistance in two arrangements and explain the results.

Core teaching point

Measure source V and total I for one resistor, two in series and two in parallel. $R_{\text{total}} = V/I$ each time; series is larger and parallel is smaller than one resistor.

Equipment and preparation

Low-voltage supply, two identical resistors, switch, ammeter, voltmeter, leads; existing Resistance worksheet pp. 5-8.

Lesson sequence

Time	Phase	Teacher move and evidence
0-6	Predict	Rank one, series and parallel by current and resistance.
6-13	Set up	Model source V and total I meter positions.
13-32	Required practical B	Measure all three layouts; swap roles.
32-43	Explain	Calculate and account for the ranking.
43-50	Check	Five-mark question; correct parallel reasoning.

Workbook task guidance

Task 1: Draw all three arrangements and predict the ranking of their total currents.

Expected: For identical resistors: series current smallest, single middle, parallel largest.

Task 2: Build each low-voltage circuit. Measure source p.d. and total current; calculate R_{total} in the table.

Expected: Use measured source V and supply I ; calculate V/I for each configuration. Exact values depend on the resistors.

Task 3: Explain why the two-series value is higher and the two-parallel value is lower than a single resistor.

Expected: Series puts opposition on the same path; parallel provides another path, increasing total I at fixed V .

Task 4: Evaluate: identify a source of measurement uncertainty and a change that would improve the comparison.

Expected: Examples: meter resolution/contact resistance; repeat readings, tight connections, same supply and temperature.

Worked example

At 6.0 V: one 10 ohm resistor draws 0.60 A; two series draw 0.30 A, so $R_{\text{total}} = 20$ ohm; two parallel draw 1.20 A, so $R_{\text{total}} = 5.0$ ohm.

AQA-style checkpoint

[5 marks] Two identical resistors each have resistance 12 ohm. Find their series resistance. A student measures 6.0 ohm for their parallel combination. Explain why this is plausible and state one control for a fair comparison.

Mark guidance: Series = $12 + 12 = 24$ ohm (2). Parallel adds a path, so more total current at the same p.d. and lower total R (2); same supply p.d./same resistors/temperature (1).

Anticipated misconception

The reciprocal formula for parallel resistance is beyond this AQA unit; understand the qualitative rule.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.

Practical hand-off

Use the workbook practical record for data and graphing. The existing full AQA required-practical worksheet supplies detailed method, circuit diagrams and additional analysis. Use low voltage and local lab safety rules.